The Future of Space Exploration in Our Lifetimes

【太空探索四大突破：太空电梯成真、小行星采矿致富、火星殖民生存、轨道数据中心崛起】

Summary: Scientists explore four plausible advancements in space technology that could become reality within our lifetimes. The concept of space elevators using materials like graphene could revolutionize orbital transport and enable interplanetary travel. Asteroid mining offers access to immense mineral wealth while potentially reducing Earth's environmental burden from resource extraction. Establishing permanent settlements on Mars presents extreme challenges but may be achievable with technological aids like robotic exoskeletons. The growing satellite network risks triggering Kessler Syndrome, though space-based manufacturing and orbital data centers could emerge as sustainable solutions.

摘要： 科学家预测本世纪可能实现的四大太空突破。基于石墨烯材料的太空电梯将彻底改变轨道运输方式，实现行星际旅行。小行星采矿不仅能获取价值数万亿的矿物资源，更有望减轻地球生态开采压力。火星永久定居点虽面临大气毒性土壤等生存挑战，但可通过机器人外骨骼等技术辅助实现。尽管卫星激增可能引发凯斯勒综合征链式反应，但太空制造和轨道数据中心或将成为可持续解决方案。

⏱️ Estimated Reading Time: 7 min.

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We can't predict the future, and when it comes to imagining the future of space exploration,

我们无法预测未来，而当谈到想象太空#探索#的未来时，

it's easy for things to get a little, well, sci-fi.

事情很容易变得有点科幻。

But what if they're not?

但如果它们并非如此呢?

Here are four things that could actually happen in our lifetimes.

以下是四件可能在我们有生之年真正发生的事。

For decades, science fiction writers have imagined a future

数十年来，科#幻#作家一直在想象这样一个未来，

where space elevators transport us almost effortlessly into orbit.

在那里，太空电梯几乎毫不费力地把我们送入#轨道#。

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Scientists now believe that this could, in theory, become reality,

科学家们现在认为，这在理论上可能成为现实，

using new materials like graphene, thought to be the strongest of all known materials.

借助石墨烯等新材料，它被认为是目前已知最坚固的材料。

Picture this.

想象一下：

A docking station.

一个对接站。

Somewhere on Earth, with an ultra-strong and super-light weight to have a cable,

在地球上的某处，设有一根超强且超轻的缆索，

which would crawl up like a lift into orbit.

它会像电梯一样#攀升#进入#轨道#。

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Space elevators could also transport things between planets and moons.

太空电梯还可以在行星与卫星之间运输物品。

That could be useful, for example, on Mars, which has a very thin atmosphere,

例如在#火星#上会很有用，因为那里大气稀薄，

making it really hard to land a spacecraft.

这使得#航天器#很难着陆。

A quick space elevator trip between Mars' moon, Phobos,

一趟快速的太空电梯，从#火星#的卫星火卫一出发，

and the Martian surface could make things much easier.

抵达火星表面，能让事情容易得多。

Space tourism could become a very real thing.

太空旅游可能会成为现实。

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You take a space elevator to the orbit and then take another mode of transport to your destination.

你乘太空电梯进入#轨道#，再换另一种交通工具前往#目的地#。

Day trip to the moon, anyone?

想来一趟月球一日游吗?

The planets, moons, and asteroids in our solar system, are packed with rare materials,

太阳系中的行星、卫星和小行星都#充满#稀有材料，

such as nickel, iron, and cobalt, which are used for batteries, electronics,

例如#镍#、铁和钴，它们被用于电池、#电子#产品，

and many low carbon technologies.

以及许多低碳技术。

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The most valuable asteroid in the solar system is 511 Davida,

太阳系中最有价值的小行星是511戴维达,

whose value is estimated at 27 quintillion dollars.

其价值估计为27万亿亿美元。

That's 27 and 18 zeros.

那就是27后面加18个零。

Meanwhile, other objects in the asteroid belt a full of diamonds.

与此同时，小行星带中的其他天体则充满了钻石。

We could potentially reduce damage to the Earth's ecosystem by shifting the mining of many resources to space.

通过将许多资源的开采转移到太空，我们可能减少对地球#生态系统#的破坏。

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Some companies and governments are already interested in exploring this, but is it theirs to exploit?

一些公司和政府已经有兴趣探索此事，但这些资源真的归他们#开采#吗?

In the near future, space mining is more likely to be used to make human populations in space self-sufficient.

在不久的将来，太空采矿更可能被用来让太空中的人类人口自给自足。

Supplying fuels and materials to space stations and settlements on the moon and Mars,

为太空站和月球及#火星#上的定居点提供燃料和材料,

much easier than taking it with us from Earth.

比从地球运去要容易得多。

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In the next 50 years, it's very possible that the surface of the moon

在未来50年，月球表面很可能

will be dominated by industrial scale mining and giant structures that we felt from the materials we might.

将被工业规模的采矿和由这些材料建造的巨大结构所主导。

In 50 years time, there may well be permanent settlements on Mars,

50年后，#火星#上可能会有永久定居点，

but life for those early explorers will be tough as no breathable atmosphere,

但早期探险者的生活将很艰难，因为那里没有可呼吸的大气，

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relentless dust storms and experience temperatures.

还要面对#无情的#沙尘暴和极端温度。

These pioneers will face lots of challenges to grow and sustain a human presence on Mars.

这些先驱者将面临许多挑战，以在#火星#上维持和#支撑#人类存在。

For example, how to grow food?

例如，如何种植粮食?

Martian soil is full of chemicals toxic to humans, meaning that any food grown there would be lethal.

火星土壤中含有对人类#有毒#的化学物质，这意味着在那里种植的食物会是#致命#的。

Humans living on Mars may need all sorts of gadgets and gizmos to help them cope in that environment,

生活在#火星#上的人类可能需要各种设备和工具来帮助他们适应那里的环境，

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such as robotic exoskeletons to counteract muscle wastage and loss in bone density.

比如机器人外骨骼，用来#抵消#肌肉流失和骨#密度#下降。

Satellites orbiting the Earth tell us the weather, provide GPS navigation and climate modelling.

环绕地球的卫星告诉我们天气，提供GPS#导航#和气候建模。

And since 2019, the number of satellites orbiting Earth has increased from approximately 2000 to more than 10,000.

自2019年以来，地球轨道上的卫星数量已从大约2000颗增加到超过10000颗。

Every time a satellite collides with a piece of space junk, it increases the chance of future collisions.

每当一颗#卫星#与一块太空#垃圾#相撞，就会增加未来碰撞的几率。

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This could set off a chain reaction known as Kessless Syndrome,

这可能会引发一种被称为凯斯勒#综合症#的连锁反应，

where the Earth is surrounded by so much space junk that launching anything into space becomes risky.

在这种情况下，地球会被太空#垃圾#包围，以致发射任何物体进入太空都变得危险。

But there's huge potential too.

但其中也蕴藏着巨大潜力。

Some chemical reactions happen differently in space.

一些化学反应在太空中的表现与地球不同。

3D printing in microgravity works really well, for example, much better than on Earth,

例如，在微重力下的3D打印效果非常好，远胜于地球，

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making it possible to create replacement organs in space.

这使得在太空中制造#替代#器官成为可能。

Global electricity demand has skyrocketed due to advances in AI.

由于人工智能的进步，全球电力需求激增。

Google's greenhouse gas emissions alone have risen 48% in five years.

仅谷歌的#温室#气体排放量就在五年内增长了48%。

In the future, power hungry data centers could be located in orbit,

未来，耗电巨大的数据中心可能会设在#轨道#上，

fed by the plentiful renewable energy from giant solar panels.

由巨型太阳能电池板提供#充足#的#可再生#能源。

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And manufacturing in space could become a thing.

太空制造也可能成为现实。

Who knows, maybe the label on your next prescription or a t-shirt might one day say made in space.

谁知道呢，也许你下一个药方的标签或T恤上，有一天会写着“太空制造”。